



KAVOSH ROBOTICS · U15 CAMP

MISSION IMPOSSIBLE

TRAINING CAMP · DAY 1

Today you will solve **2 robot missions** and **1 hands-on mission**. Nobody knows the exact challenge until your coach reveals it — just like a real competition.

KAVOSH ROBOTICS — U15 MISSION IMPOSSIBLE CAMP

SUPPLY DROP

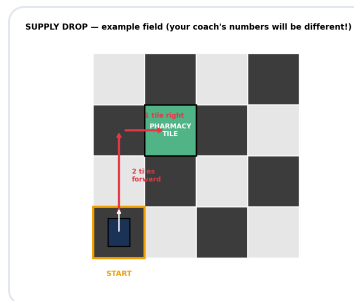
A village is cut off after the earthquake and desperately needs medicine. Your robot must cross the broken field, find the hidden Pharmacy tile, and drop off a rescue pack – using nothing but counted tiles to find the way.

YOU'LL NEED

- SPIKE robot (down color sensor)
- Checkerboard field
- 1 rescue cube

HOW IT WORKS

- 1 Your coach secretly picks a Pharmacy tile and writes down directions, like “2 tiles forward, 1 tile right.”
- 2 You get 2 minutes to type those two numbers into your program. You do NOT get to see the tile.
- 3 Press start. Your robot must count tiles as it drives – and stop exactly on the Pharmacy tile.
- 4 Drop your rescue cube on that tile.
- 5 You have 90 seconds total. You get 2 tries – your best try counts.



HOW YOU SCORE

What happens	Points
Robot stops exactly on the Pharmacy tile	60
Robot stops next to it (1 tile off)	30
Rescue cube ends up on the tile	+30
You finish in under 60 seconds	+10

Best possible score: **100 points**

THE LONG CORRIDOR

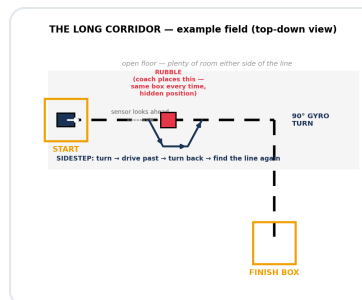
The rescue tunnel is long, and somewhere along it the ceiling has collapsed. Your robot cannot push through – it must sense the rubble, stop, calmly step around it, and find its way back to the path. Then it follows the tunnel around the corner and out the exit.

YOU'LL NEED

- SPIKE robot (down + front sensors)
- Taped line + open floor on both sides
- A small box (same one every round)

HOW IT WORKS

- 1 Your coach places the rubble box somewhere along the straight part of the line. You will not know exactly where – but it's always the same box, in the same orientation.
- 2 Press start – your robot follows the taped line down the corridor.
- 3 Your front sensor must notice the rubble and stop before touching it.
- 4 Your robot then does the same sidestep move every time: turn away from the line a set amount, drive forward far enough to clear the box, turn back, then drive until it finds the black line again.
- 5 Once back on the line, follow it down to the corner, turn 90° using the gyro sensor, and follow the last stretch of tape into the Finish Box.
- 6 You have 2 minutes. You get 2 tries – your best try counts.



HOW YOU SCORE

What happens	Points
Robot stops before touching the rubble	25
Robot bypasses it and finds the line again	35
Robot makes the 90° turn	20
Robot reaches the Finish Box	20
Robot touches the rubble	lose the 25, keep going

Best possible score: **100 points**

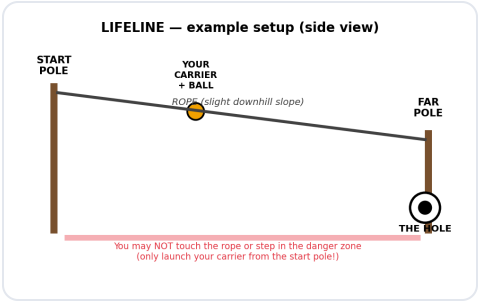
LIFELINE

A rescue line has been strung across the danger zone. A tiny medicine capsule (a ping-pong ball) must travel from the start pole, all the way across, and drop into the hole at the far end. You may not throw it, carry it by hand, or step into the danger zone.

YOU'LL NEED

- 1 rope (~2 m)
- 1 ping-pong ball
- Cup + string + tape (limited amounts)

- HOW IT WORKS**
- 1 Using only your allowed materials, build a small carrier that can ride along the rope.
 - 2 Load the ball onto your carrier at the start pole.
 - 3 Let it travel along the rope to the far end – you decide how (gravity, a push, a pull-string).
 - 4 The ball must land in the hole at the end.
 - 5 Once the ball is loaded, you may not touch the rope or step into the danger zone – only launch your carrier from the start pole.
 - 6 You get 3 tries, 3 minutes each. Your best try counts.



HOW YOU SCORE

What happens	Points
Ball (or carrier) reaches the far pole	30
Ball lands in the hole	+50
Fastest successful run in the room	+10

Best possible score: 90 points

Day 1 – Quick Summary

Mission	Type	Difficulty	Main Skill
Supply Drop	Robot	Medium	Tile-counting navigation
The Long Corridor	Robot	Hard	Obstacle stop + sidestep + gyro turn
Lifeline	Hands-on	Medium-Hard	Simple mechanisms & teamwork

Coach's plan for the day: Supply Drop (morning) → Lifeline (after lunch, shorter & fun) → The Long Corridor (afternoon).